# Thesis Abstract: Improving Multi-Object Navigation in an LLM-Based Embodied AI Framework

(Full thesis available on *GitHub* )

## Introduction

Large language models have shown remarkable abilities in planning, reasoning, and solving complex tasks by generating code. In recent work such as ViperGPT[1] and VisProg[2], LLMs are used to generate complex compositional programs for language-vision-tasks like object detection or visual question answering, allowing them to solve complex visual tasks without any training. These systems show significant results in vision-language challenges just by using a few in-context examples to create complex programs by combining different modules without training. Figure 1 illustrates how VisProg leverages an LLM to generate multi-step visual programs from natural language prompts.

Embodied AI[3] focuses on building agents that can move, see, and reason within 3D environments essentially learning through interaction with their environments rather than relying solely on static data. Embodied AI has seen significant progress in recent years, largely thanks to the development of fast, photorealistic simulators [4, 5]. These tools allow researchers to run large numbers of parallel simulations in realistic indoor environments, making it possible to test and evaluate agents at scale something that was much harder in traditional robotics setups.

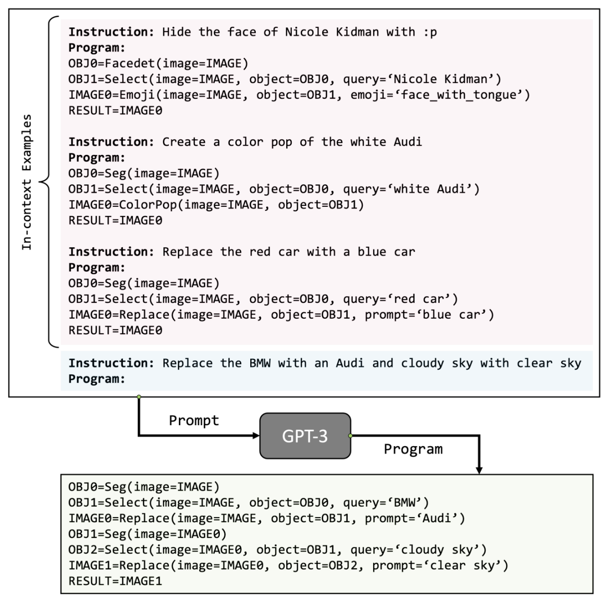


Figure 1: Example from VisProg examples [1].

## TANGO Framework

TANGO[6] is a framework in embodied AI that leverages LLMs' capabilities to generate programs; this work seeks to extend this ability to embodied agents, enabling them to observe and control their actions in zero-shot navigation rather than learning task-specific policies. The complete set of TANGO modules, including navigation, vision-language, and functional subroutines, is illustrated in Figure 2.

The agent receives a natural language prompt (e.g., "Is the shower curtain open?"), The LLM generates a structured program composed of high-level commands like detect, navigate-to, or answer and integrates it into the Habitat-Sim framework[7].

As shown in Figure 3, the LLM is guided by a fixed set of in-context examples, which act as a prompt template. It depicts how to break a complex task into smaller steps.

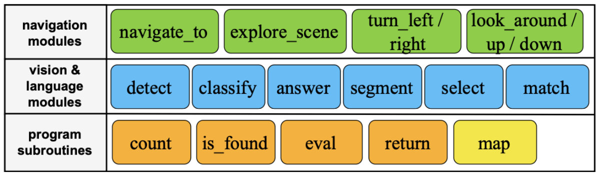


Figure 2: Overview of TANGO modules.

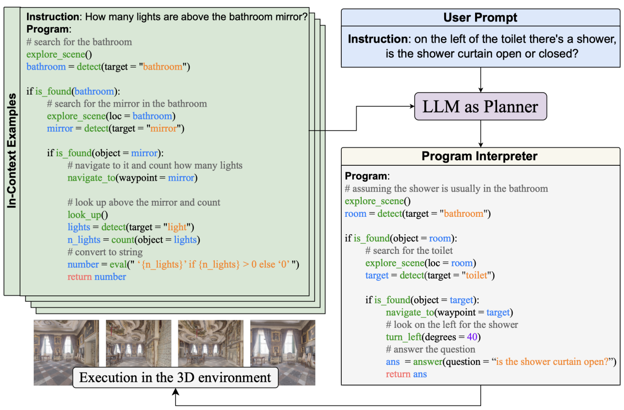


Figure 3: Overview of program generation and execution in TANGO.

The generated program is passed to the TANGO's program interpreter, which maps the generated code to each module for execution; the LLM only generates high-level planning, and the program interpreter executes it. For example, the detect() is mapped to the whole detect module, executes the full code, and returns the output. In this way, I can detect failures in specific modules and enhance them. This design allows TANGO to generalize across a wide range of embodied challenges.

# Enhancing TANGO Detection for MultiON Tasks

The target of this thesis is to enhance a bottleneck in TANGO; for some tasks, the accuracy and success rate are low due to failures in the detection part, specifically in MultiON[8] tasks. The baseline open-set detector, OWL-ViT2[9], showed poor performance in these cases, prompting a focused effort to improve the detection pipeline.

I tackle this problem by fine-tuning several state-of-the-art detection models, including YOLOv8 [10] and DETR [11] variants on our synthetic dataset for MultiON task using Habitat-Sim[12] and HM3D[13] scenes. I got an improvement by using our best-performing model, YOLOv8-X, in the detection module.

When evaluated on the MultiON 3-ON task, the upgraded detector improved the agent's success rate by 28% compared to the OWL-ViT2 baseline. It also outperformed the MOPA framework [14] by 14.6% in success and exceeded the best end-to-end model, ProjNeuralMap [8], by 26%. These results depict the point of integrating fine-tuned detectors in the detection module. Despite this limitation, the TANGO-based approach achieves significantly excellent results, demonstrating its effectiveness in generalizable and zero-shot navigation in MultiON without retraining.

| Model | Success (%) | Progress |
|---|---|---|
| PredSem [14] | 38 | 53 |
| No-Map [14] | 10 | 24 |
| ObjRecogMap [14] | 22 | 40 |
| ProjNeuralMap [14] | 27 | 46 |
| YOLO8n With Memory | 48 | 67 |
| YOLO8m With Memory | 49 | 69 |
| YOLO8l With Memory | 47 | 67.3 |
| **YOLO8x With Memory** | **52.6 (+28%)** | **70 (+27%)** |
| **Detr(resnet50)** | **36(+12%)** | **57(+14%)** |
| **Baseline (Owl-ViT2) With Memory** | 24 | 43 |
| YOLO8x Without Memory | 49 | 68.8 |
| Baseline (Owl-ViT2) Without Memory | 19 | 39 |
| Baseline (Owl-ViT2) Without Memory | 19 | 39 |

Table 1: Performance Comparison on MultiON 3-ON Task

Additionally, ablation studies on TANGO's memory mechanism are conducted to understand its power in agent performance by creating a map and remembering the visited areas instead of searching for objects. The results confirmed improvements across both 3-ON and 2-ON settings when memory was used in combination with the upgraded detector. I summarize the performance of various YOLO variants and baselines on the MultiON 3-ON task in Table 1.

In summary, this work demonstrates that using improved object detectors and a memory mechanism significantly boosts the performance of the MultiON task. By leveraging large language models for program generation, the system achieves robust zero-shot generalization in embodied AI.

# References


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